DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 1 Winter 2025 (Regular)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 1

Subject Code & Name: 24AF1000BS101 - ENGINEERING MATHEMATICS – 1

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.

2.Question No. 1 will be compulsory and include objective-type questions.

3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6

4.Use of non-programmable scientific calculators is allowed.

5.Assume suitable data wherever necessary and mention it clearly.

20 marks

(11)

Q1. Objective type questions. (Compulsory Question) 12

1 A homogeneous system of linear equations AX = O has infinitely many solutions if.....

a) rank (A) = number of unknowns

b) rank (A) ≠ number of unknowns

c) determinant is non-zero

d) none of these

2 The eigen values of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ are.....

a) 2, 0, 0 b) 2, 3, 0 c) 2, 3, 4 d) 2, 3, 5

3 The eigenvalues of a matrix A and its transpose (A^T) are.....

a) different b) negative of each other

c) same d) reciprocal

4 If $u = x^3 + y^3$ then the value of $\frac{\partial^2 u}{\partial x\, \partial y} =$

a) $3y$ b) -3 c) 3 d) 0

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5 If $z = f(x, y)$ and $x = \phi(u, v)$ and $y = \psi(u, v)$ then which of the following is correct?

a) $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$ b) $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{dx}{du} + \frac{\partial z}{\partial y}\frac{dy}{du}$

c) $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} - \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$ d) none of these

6 If $u = \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$ is homogeneous function of degree

a) $\frac{1}{2}$ b) 0 c) 1 d) $\frac{-1}{2}$

7 The necessary conditions for function $f(x, y)$ to be maximum or minimum at a point (a, b) is.....

a) $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$ b) $\frac{\partial f}{\partial x} > 0, \frac{\partial f}{\partial y} < 0$

c) $\frac{\partial f}{\partial x} < 0, \frac{\partial f}{\partial y} > 0$ d) none

8 If u, v are the function of r, s and r, s are the functions of x, y then $\frac{\partial(u,v)}{\partial(x,y)}$ =.....

a) $\frac{\partial(u,v)}{\partial(x,y)}\frac{\partial(x,y)}{\partial(r,s)}$ b) $\frac{\partial(u,v)}{\partial(x,y)}\frac{\partial(r,s)}{\partial(x,y)}$

c) $\frac{\partial(u,v)}{\partial(r,s)}\frac{\partial(r,s)}{\partial(x,y)}$ d) none of these

9 The polar curve passes through the pole when.....

a) $r = \infty$ b) $r = 0$ c) $\theta = 0$ d) $\theta = \pi$

10 The asymptote to a cartesian curve parallel to the X-axis, if it exists, is obtained by equating to zero the.....

a) highest degree term in the equation

b) lowest degree term in the equation

c) coefficient of the highest degree term in x

d) coefficient of the highest degree term in y

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11 The value of $\int_0^1 \int_0^x y^2\, dx\, dy$ is.....

a) $\frac{1}{12}$ b) $\frac{1}{6}$ c) $\frac{1}{3}$ d) none of these

12 The value of $\int\int dx\, dy$ is.....

a) perimeter of region b) volume of region

c) area of region d) none of these

Q2. Solve the following. 6

A) Reduce the following matrix to normal form and hence find its rank (6)

$$A = \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$ Rank - 3

B) Test for the consistency and solve: 6

$2x - y - z = 2;\ x + 2y + z = 2;\ 4x - 7y - 5z = 2$

Q3. Solve the following.

A) If $u = \log(x^2 + y^2 + z^2)$ then prove that, 6

$$(x^2 + y^2 + z^2)\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 2$$

B) If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$ then show that, 6

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{-\sin u \cos 2u}{4\cos^3 u}$$

Q4. Solve Any Two of the following. https://www.batuonline.com

A) If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$ then show that 6

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$$

B) Expand $f(x,y) = \cos x \sin y$ in the powers of x and y as far as the terms of third degree 6

C) Examine the function $f(x,y) = x^3 + y^3 - 3axy$ for maxima and minima where $a > 0$ 6

Q5. Solve Any Two of the following.

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A) Evaluate $\int_0^a x^2(a^2 - x^2)^{\frac{3}{2}}\, dx$ 6

B) Trace the curve $y^2(2a - x) = x^3$ 6

C) Trace the curve $r = a\ sin2\theta$ 6

Q6. Solve Any Two of the following.

A) Evaluate $\int_0^1 \int_0^y \int_0^{xy} xy^2z^3\, dx\, dy\, dz$ 6

B) Change the order of integration and evaluate $\int_0^\infty \int_0^x x\, e^{\frac{-x^2}{y}}\, dx\, dy$ 6

C) Verify the Cayley-Hamilton theorem for the matrix and hence find A^{-1} 6

for the matrix $A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & -2 \end{bmatrix}$ 3

*** End ***

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